Institute of Technology, Chennai. They were declared global winners of the first edition of the Intel AI Global Impact Festival. One of these students, Rishikesh Amit Nayak, hails from a village in Orissa called Khatuapada, which lost 54% of its harvest in 2017 due to a microbial attack, and many farmers in the village were pushed into heavy debt.

Microbial and pest attacks account for 80% of crop failure in India, with the monetary value of crops lost exceeding US$500 billion. So, together with two of his friends Nayak set about finding a solution for this problem. Their solution uses thermal imagery, powered by IoT, Intel processors, and sensors, to detect microbial and pest attacks, and estimate irrigation in the fields.

Their device functions on real-time data to detect bacteria, fungi, and viruses in crops, and uses IoT to arrive at a micro-level solution," says Khurana. "Currently, the prototype version is available for ₹1,500. These students are now working with agricultural departments to validate and implement this project so that it can serve its larger societal purpose."

A 14-year-old from the Lalitpur district of Uttar Pradesh, Nandani Kushwaha, recently secured a place amongst the top 20 young innovators of the country for her AI based innovative project—*Mitti ko Jaano, Fasal Pehchano*. "This soil analysis and crop recommendation technology is a smart data based AI tool that identifies the most suitable crop that can be planted by analysing the various nutritional components present in the soil. This model can be deployed as a physical soil-detecting tool that will help farmers understand their soil accurately and make informed decisions," explains Khurana.

Likewise, Isha Biswas, a student from a Kendriya Vidyalaya school in Arunachal Pradesh, has developed an AI technology for farmers to identify crops that can be best harvested at a given time. Her solution also suggests suitable crops that can be planted in a particular type of soil after assessing several ecological and environmental factors.

'Weed and Crop Detection System' developed by Vaibhav Dewangan and Dhiraj Kumar Yadaw from Chhattisgarh is another AI powered technology, which helps identify different types of weeds in farms and aid in their removal.

We also found a couple of interesting farming bots developed by young students mentored by Dimple Verma of WhizRobo, finding a mention in the India Book of Records. Viraj Gupta, an 11-year-old boy from Ludhiana, has developed Krishi-Bot, a robot that can help with multiple tasks on a farm, from ploughing and planting to irrigation and harvesting. It can also check the soil to decide whether it needs watering or not. The robot can be controlled remotely by the farmer, and allows remote monitoring of weather. The prototype was built at a total cost of just ₹6,000.

The agricultural drone developed by Aryaman Verma can help with irrigation, spraying, and surveillance of the farms with a battery backup of 15-20 minutes. It is trained to return to the base once the battery is discharged. Equipped with cameras and sensors, the drone can also gather data on crop health, yield, and other important metrics.

Another young techie, Yashvardhan Singhvi, has made an autonomous crop-scouting and weeding robot. It is equipped with navigation systems and precision tools to carefully move around fields and remove weeds without damaging crops.

Reaching the grassroots—infra, trust, and affordability

Despite the availability of quality agri-tech solutions, why is the uptake of tech still low at the grassroots level, and how can this be overcome?

"The current situation shows that early adoption of emerging technology has only been done by larger farmers with sufficient funds and scope to afford and implement the tools for precision and smart farming. However, there are pilots and lighthouse instances where groups of small and marginal farmers at the village or district level have come together to practice smart farming, which can be scaled with state interventions," says Khurana of Intel.

Jalan suggests that these challenges can be addressed if agri-tech companies adopt a B2B or B2G model. "Government and private sector companies work with farmers and provide them with technical support, and finance to run the farm effectively. This is exactly what we do at Cropin. Farmers do not have to pay for our solutions as we deploy them in partnerships with our B2B and B2G customers, breaking the financial barrier in tech adoption. This not only solves the problem of affordability, but also ensures impact at scale, which is difficult to achieve working with individual farms holdings," he says, adding that, "Meaningful and enduring transformation of agriculture is not possible unless smallholder farmers at the grassroots level are trained and enabled to adopt smarter, more efficient, and sustainable ways of farming. Digitisation and intuitive, inexpensive, and easily accessible technology can go a long way in making this happen."

According to Khurana, we also need creative public-private partnership initiatives in India to accelerate the adoption of technology in agriculture. She cites the example of Agristack, which is based on the India Digital Ecosystem of Agriculture (IDEA) framework. "It provides data